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Deep fakes as a tool of political advertising. Can regulatory framework benefit from the "Ship of Theseus" paradox?☆,☆☆

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ABSTRACT

Deep fakes have been consistently perceived as a threat to social and political processes, particularly elections. A rarely discussed phenomenon, though gaining importance considering the increasing number of real-life use cases, is using deep fakes for political advertising purposes. By manipulating visualizations, artificial intelligence (AI) allows politicians to eliminate or minimize the deficits of their performance and highlight or create features perceived as attractive by the audience. Such manipulations pose a significant epistemic problem, and can severely distort perception of real politicians and their characteristics. These disruptions might correspond to the classic "Ship of Theseus" paradox that addresses the problem of changing identity. The aim of this study is to analyze how this philosophical conundrum could be used for crafting better transparency obligations (disclaimers) for using deep fakes for political advertising purposes. The aim of such disclaimers is to sensitize and protect public opinion against manipulation. However, at the moment the basic labels concentrate on technological dimension of manipulation, overlooking cognitive consequences for human audiences. This paper proposes extensive disclaimers to strengthen their informative and sensitizing potential.

1. Introduction

Over the centuries, the art of political advertising has been creatively developed, taking into account new forms of conveying messages, often adjusted to the appropriate audience (Langguth et al., 2021). The scope of the interference was gradually implementing technological progress and the use of mass media, making persuasion, but also misleading, manipulation, lying, or even outright falsification more refined (Gordon, 2024; Jeangene Vilmer, Escorcia, Guillaume, & Herrera, 2018; NSA, 2023; Vazquez Llorente, 2024). The recent gains in the field of generative artificial intelligence (AI), enhanced by access to datasets and computing power, can be seen as "the next revolution" in facilitating sophisticated digital manipulation of political nature (Farid & Schindler, 2020, p. 13).

Deep fakes, one of the flagship examples of generative AI capabilities, have been consistently assessed as a threat to the integrity of information, the media, electoral processes, and, consequently, democracy (among others: Farid & Schindler, 2020; Vaccari & Chadwick, 2020; Huijstee van et al., 2021). Regularly reported cases of misusing AI-generated or AI-manipulated audio or visual content for political purposes must raise justified concerns, and in many cases their outcome can be unambiguously labeled as harmful, especially when they are used for spreading disinformation about third parties.

However, deep fakes do not necessarily have to be directed against third parties to influence recipients' perception and behavior. Politicians can consensually create their AI-generated representations that resemble reality and, at the same time, significantly modify it. It is enough that the characteristic features of the depicted politician are changed. AI-powered visualizations can be used to reduce the limits of one's performance, completely get rid of attributes identified as unattractive, or add additional ones recognized as appealing (Pashentsev, 2023). Simultaneously, they make it impossible to recognize what is true, disturbing the boundaries between real and fake and undermining perception of politicians' real identities.

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Deep fakes political advertising has great potential to become a model for future political campaigns (Łabuz & Strnad, 2025). The first cases of using AI to create a new, better, modified image of politicians were registered in India, South Korea, the USA, or Sweden. Politicians are reaching for technological innovations to refine their image and adapt it to the expectations of voters, which Yoon Suk-yeol used to generate interest in younger age groups during the presidential election in South Korea in 2022. In 2024, in India, politicians of various parties generated their AI-powered avatars speaking different dialects and languages to reach linguistic minorities with their messages.

New forms of technology application require a response at the legislative level. The aim of this study is to propose enhanced labeling for deep fakes used for political advertising purposes. Basic disclaimers informing viewers about the use of AI to create the content cannot sufficiently sensitize and protect them against such sophisticated manipulation. We point to the need to expand standard disclaimers to account for various levels of manipulation, including technical, ontological, and epistemic ones. Therefore, our study should broaden the conceptualization of countermeasures against deep fakes in the form of enhanced transparency obligations.

For this purpose we refer to the "Ship of Theseus" paradox. This ancient conundrum refers to the idea of gradually replacing planks of the wooden ship and poses the question, whether these changes result in an emergence of a completely new object. We compare this with changing the characteristics of politicians by deep fakes and thus distorting the image and perception of their identity. The "Ship of Theseus" paradox is employed here as a conceptual framework to illustrate the challenges and consequences of deep fakes political advertising, both in ontological and epistemic dimensions. While the paradox traditionally examines material continuity (Heller, 1987), its significance in this study lies in demonstrating how AI-powered modifications violate faithful representations.

In light of the increasing use of deep fakes for political advertising purposes, deep fakes political advertising requires a revised approach and greater involvement from policymakers and regulators. In this context, we aim to create a theoretical framework for better disclosure rules, proposing specific wording of extensive disclaimers.

Therefore, the overarching research questions of the study are:

RQ1: *How deep fakes can be used to change specific features of the politicians to increase their attractiveness?*
RQ2: *Can the "Ship of Theseus" paradox be useful in recognizing possible safeguards for deep fakes political advertising?*
RQ3: *How to formulate transparency obligations to take into account the multi-dimensional manipulative potential of deep fakes political advertising?*

2. Deep fakes as a tool for manipulation

Deep fakes might be defined as AI-generated or AI-manipulated audio, image or video content that appreciably resembles existing persons, objects, places or other entities or events and would falsely appear to a person to be authentic or truthful (European Commission, 2024). The very definition determines the manipulative or deceptive nature of deep fakes due to the manner of their creation on the basis of already existing content (Łabuz, 2023; NSA, 2023) and the outcome of the depiction. They present a false reflection of reality, pretending to be authentic or truthful and thus misleading recipients (Boté-Vericad & Vállez, 2022).

Numerous authors have consistently analyzed the harmful potential of deep fakes, warning against their misuse and far-reaching social and political consequences (Chesney & Citron, 2019; Europol, 2022; Farid & Schindler, 2020; Huijstee van et al., 2021; Kleemann, 2023). Non-consensual malicious deep fakes are gaining importance as a form of political warfare, although their impact on the course of the elections and political processes is difficult to quantify (Łabuz & Nehring, 2024). Deep fakes are already used for cybercrime and phishing schemes, creating non-consensual intimate images, or Child Sexual Abuse Materials (Huijstee van et al., 2021; Europol, 2022; Łabuz et al., 2025).

One of the significant threats related to deep fakes is identity theft (Agarwal & Ratha, 2023; Europol, 2022; Mirsky & Lee, 2021), the process of impersonating other people, often for the purpose of fraud. Öhman (2022) devotes detailed considerations on the epistemic limits of identity. Appropriate matching with a particular person's image is normally based on visual identifiability or recognizability. In the future, such phenomena will require increased involvement on the part of law enforcement, significant investments in detection technologies, as well as a rethinking the approach to image protection. Deep fakes are already being used to digitally revive the dead, also for commercial purposes (Pawelec & Łabuz, 2025). Although such solutions can have positive potential (e.g. in education, culture or cinematography), they also carry significant threats to image integrity and the potential for abuse.

Some researchers (Huijstee van et al., 2021) point to the cascade of negative social and political consequences that might accompany the occurrence of deep fakes in the information space, whereas Europol (2022) estimates that the majority of deep fakes can be attributed to the malicious intent of their creators or disseminators. Due to the prevalence of these negative applications, researchers mostly concentrate on deep fakes created to harm third parties.

On the other hand, deep fakes find many unequivocally positive uses, as they can be employed for leisure or scientific purposes, including education, psychology, or healthcare (Farid & Schindler, 2020). Discussing them in detail is beyond the scope of this study. However, one should warn against excessive demonization of technology, which in some cases could even result in constructing disproportionate regulatory frameworks.

For the time being, some cases defy simple binary classification. The topic of improving one's performance for political advertising purposes is under-researched. In this study, we focus on discussing only those forms of political advertising with the use of deep fakes that are aimed at improving the image of politicians by changing the characteristic features of their performances or adding specific attributes they do not have (Łabuz & Strnad, 2025).

Since the appearance of deep fakes in the information space in 2017, it has been predicted that they will lead to a decline in the epistemic value of audio and visual materials (Fallis, 2021; Rini, 2020) or threaten the normative claim that *seeing means believing*. This belief should be seen as an emanation of *ocularcentrism* created over the years of experience with the media recognized as reliable (Geddes, 2021). Although these disruptive phenomena have not reached the level of the predicted "epistemic apocalypse" (Fallis, 2021; Łabuz & Nehring, 2024), recent studies show a high level of uncertainty about the credibility of the media, a low level of trust in them, and increasing problems with distinguishing truth from falsehood among the citizens, which is directly and indirectly associated with deep fakes' potential (Home Security Heroes, 2023; Twomey et al., 2023). Other information disorders, such as disinformation, have also contributed to consolidating low trust in news. Deep fakes play the role of a catalyst for negative phenomena in the information space and contribute to the post-truth narratives that further distort the common understanding of reality. At this point visual media still carry the epistemic value, and the above-mentioned gradual disintegration has not led to complete epistemic breakdown, as we still use visual representations to shape our perception of reality (Smith, 2024). However, Fallis (2021, p. 625) suggests that "as deepfakes become more prevalent, it may be epistemically irresponsible to simply believe that what is depicted in a video actually occurred".

The phenomena described above will lead to an even greater distortion of reality and questioning of its common understanding. The assessment of the epistemic value of deep fakes should also take into account the long-lasting effects of their appearance and display (Fernandez, 2021; Nagumotu, 2022). Exposure to fake content can cause irreversible consequences for cognitive abilities and influence the

autonomy of decision-making processes by strengthening cognitive biases (Brown, 2020). Especially since each deep fake bears an element of manipulation in the technological layer, as deep fakes constitute manipulated media items (Yadlin-Segal & Oppenheim, 2021). Harris (2024) distinguishes different types of threats stemming from deep fakes, including deception, growing skepticism, and finally the disintegration of epistemic value. These epistemic threats interpenetrate each other, and must be included in the discourse on the use of deep fakes for political advertising purposes.

3. Political advertising employs technology

Political advertising can be understood as a tool of communication used by politicians and political parties aimed at persuading recipients, so they take favorable actions, to eventually win the support and votes of the electorate (Olalekan & Sam, 2018). Historically it has been the subject of numerous studies, which focused mainly on influencing electoral preferences, traditional forms of conveying messages, and more recently on a decay in their effectiveness, and the mass use of the Internet and social media (e.g., Coppock et al., 2020; Franz & Ridout, 2010; Kahn & Geer, 1994; Tanusondjaja et al., 2023). Due to its significance for shaping political and social reality, it is a subject to special protection regimes introduced mainly in national electoral codes. The EU *Regulation on the transparency and targeting of political advertising* aims to create a generic regulatory framework that is supposed to answer at least some of the technological challenges, and might serve as a benchmark for further work (European Commission, 2024b). It also contains a legal definition of the term "political advertising", aiming for harmonization of the rules among the EU Member States. Therefore, "political advertising" is seen as the preparation, placement, promotion, publication or dissemination, by any means, of a message: (a) by, for or on behalf of a political actor, unless it is of a purely private or a purely commercial nature; or (b) which is liable to influence the outcome of an election or referendum, a legislative or regulatory process or voting behaviour (European Commission, 2021b). This definition reflects the general idea of what political advertising is, focusing on the subject and form of undertaken activities, the responsible entities, and the main goals and context.

Surprisingly, this regulation lacks references to deep fakes (although it mentions general technological developments), which means that the relevant provisions can only be derived on a subsidiary basis. The failure to specify or directly address a specific problem in regulatory framework may be a conscious decision that allows for the future interpretation of specific solutions from general provisions. Drawing attention to the problem of technological development gives hope for additional regulatory guidance at some point.

It might also result from the relative novelty of the solutions described in this study as *deep fakes political advertising*. By this term we understand the actions aimed at using deep fakes to strengthen one's own political advertising activities, also by eliminating or minimizing certain features that politicians have, or adding those that they do not have, to increase their attractiveness among the recipients to influence political behavior (Łabuz & Strnad, 2025).

The development of political advertising techniques comprised a number of historical technological revolutions, including the use of radio by Franklin D. Roosevelt, TV by John F. Kennedy, and social media by Donald Trump (Gordon, 2024). Each of them allowed for an increase in the number of recipients, indirectly contributing to greater inclusiveness and social participation. The technological revolution has enabled new trends of *digital* or *online political advertising* (Heinmaa, 2023; Jaursch, 2020; Tanusondjaja et al., 2023), contributing to the voluntary and involuntary political engagement of citizens, mainly due to the time they spend on the Internet and social media, or the growing number of interactions with political content (Koc-Michalska et al., 2016; Neudert & Marchal, 2019; Vaccari & Valeriani, 2021). Low entry barriers allowed politicians to intensify their activities, accelerate the exchange of information, and change the parity of campaign spending in favor of online campaigns (Neudert & Marchal, 2019; Zhuravskaya et al., 2020). The ease of creating and disseminating online political advertising shortened the virtual distance between politicians and citizens, replacing at least some of the traditional interactions in the form of election rallies (Dommett, 2019). Online campaigning weakened the role of traditional media (Gordon, 2024), and shifted the focus to presenting further decontextualized fragments of reality.

Political campaigns draw heavily on the experience of classic marketing. They are often based on mere ideas of what reality looks or might look like. They might create just "a desired image or impression in the minds of consumers, rather than reflect any underlying truth or reality" (Nosta, 2023). Therefore, it seems right to say that "we are the objects of countless influence campaigns and we understand some of them more than others", becoming particularly vulnerable at moments when advertising, regularly associated with manipulation, is just irresistible (Susser et al., 2019, p. 10). These moments, however, are eminently discretionary. In the realm of politics, the relationship between voter's personal preferences and candidate's abilities involves many factors in a constantly changing political and social environment. Researchers focus on "consideration sets", estimating that voters would "concentrate on the individuals corresponding to relevant attribute groups", which obviously includes party affiliation and political beliefs, but also specific features assigned individually to the candidate (Järvi et al., 2021, p. 444).

The subtle difference between advertising, persuasion, and manipulation is often difficult to grasp, and the boundaries are not directly defined (Abuelwafa, 2021; Crain & Nadler, 2019; Jones, 2021; Noggle, 2021). Manipulation can be seen as an attempt to influence one's behavior by intruding on autonomous decision-making processes, also by means of deception (Mills, 1995; Whitfield, 2022). The deception, coercion, or the covert manner do not necessarily have to be applied (Whitfield, 2022), though they might enhance the outcome of manipulation. Although some authors (Jones, 2021) see the hidden nature as an essential element, in our opinion, it is not fully consistent with the development of new forms of manipulation with the use of modern technologies. In the case of deep fakes, already blurred boundaries between advertising and manipulation become even more blurred, and manipulation extends to new epistemic layers. Susser et al. (2019) use a colloquial but very accurate term, "sneaky", when the goals of advertising are achieved covertly. Deep fakes can be seen as a perfect example of combining covert and non-covert forms at the same time, which in this case makes it hardly possible to draw the line between manipulation and advertising, or set the clear limits of admissibility.

4. Deep fakes as a tool of political advertising

Through AI synthesis, deep fakes allow for significant interference with the original audio and visual media by modifying or replacing some parts of the recording, or even their production solely by AI. Based on previously recorded materials, AI learns to duplicate the image or its parts. Using the so-called generative adversarial networks, consisting of two elements – a generator and a discriminator – AI verifies whether the presented image can be detected as false and learns how to improve it by synthesizing each frame, and reaching realistic results (Walczyna & Piotrowski, 2023). Recent research shows that advances in technology make it impossible for most people to identify good-quality deep fakes (Krueger et al., 2023), whereas the real and fake faces are already indistinguishable (Nightingale & Farid, 2022).

Machine learning allows for a great intervention and elimination or addition of specific features, as well as equipping the synthesis object with different ones. Transferring this to the field of deep fakes political advertising, politicians can use these tools to exclude features perceived as unattractive, and highlight elements that will increase their attractiveness (Łabuz & Strnad, 2025). They may seem more energetic, smiling, and relaxed. They can completely change their body language.

Or generate a credible speech in a foreign language they do not speak (Łabuz & Nehring, 2024). By using advanced communication mechanisms, they can create an AI-representation, so-called *avatar*, and engage it in the interactions with the recipients, which gradually includes real-time interactions (Butala et al., 2025). And all this quickly and effortlessly through AI-synthesis, though these processes need significant computing power.

Until the beginning of 2025, the first concrete cases were recorded around the world – differing in scale and method of implementation – which shows the practical applications of deep fakes political advertising (Łabuz & Strnad, 2025). Already in 2020, deep fakes political advertising, though it was not a part of the official campaign, was used to enhance the performance of Indian politician Manoj Tiwari. His speech was translated into the Haryanvi Hindi dialect he does not normally speak, which allowed him to reach additional groups of voters. AI visualized Tiwari's speech, creating the impression that he actually spoke this dialect (Nilesh, 2020). The main target of the campaign was linguistic minorities excluded from the mainstream political discourse in India.

The turning point in the implementation of a coordinated campaign using deep fakes political advertising was the actions taken by the staff of Yoon Suk-yeol, the candidate of the People Power Party, in the presidential elections in South Korea in 2022. Based on hundreds of previously recorded videos, the AI generated an avatar of the politician, which at first glance looked and sounded like the real Yoon (Martin & Yoon, 2022). However, AI Yoon did not reflect the candidate's characteristic features. He was relaxed, joked regularly, and used the slang of the young generation. Some observers pointed out that some of Yoon's public speaking deficits were eliminated through synthesis, but this may have been a coincidence rather than a deliberate attempt to increase attractiveness. Regardless, the not-so-beneficial habit of shaking his head while speaking was allegedly much less noticeable (Kim, 2022). The election staff allowed the public to ask questions, to which AI Yoon's responses were then crafted by the election team and generated in the form of audiovisual deep fakes by AI. The main target of the campaign was the younger generation of voters that massively shared the content, allowing the People Power Party to reach dozens of millions of views and interactions (Halvarsson, 2024; Shin & Yi, 2022). The case of AI Yoon has been pointed out by researchers as an example of the successful transposition of video deep fakes for political advertising, or a potential source of success of the candidate (Farid, 2022; Martin & Yoon, 2022; Pashentsev, 2023; Shin & Yi, 2022).

Similar solutions were also used in the USA. In 2023, Francis Suarez, running in the Republican Party primaries, created his own avatar that was supposed to answer voters' questions. However, the campaign did not have a mass character, and Suarez himself withdrew from the elections (Harrison, 2023). The reach of the campaign implemented in 2023 by New York Mayor Eric Adams, who used audio deep fakes to make automatic phone calls (so-called *robocalls*) to people representing linguistic minorities, is unknown. Although Adams himself does not speak Yiddish or Mandarin, his deep fake representation made statements in both languages while calling (Simon, 2023). The actions were criticized by commentators, and Adams defended himself with the need to communicate with the city's residents (Izaguirre, 2023). In Sweden, the representative of the Sweden Democrats Party tried to reach selected communities by synthesizing original statements in Arabic (Halvarsson, 2024).

Each of these cases is characterized by a different scope of AI application and, therefore, a different degree of manipulation. Undoubtedly, in each case, the image of the presented politician was not consistent with reality, although the distortions concerned various areas of their performances. The AI-alter egos created in this way enabled political advertising.

It is worth noting that deep fakes political advertising in this form had a mobilizing function and, under certain conditions, can counteract political exclusion. The example of India is perhaps the most prominent one because it refers to linguistic diversity that prevents a significant part of the population from democratic participation (Pawelec, 2022). Characterizing deep fakes political advertising should therefore take into account diversified factors, even if the primary goal is not to strengthen participation or deliberation but to attract voters. However, the line is extremely thin, and even a positive goal does not change the fundamental problem of manipulation.

A legitimate question might be raised: whether the use of deep fakes is fundamentally different from using traditional tools of media manipulations applied also for political advertising purposes (e.g., photo editing, including photoshopping)? Generally, they are not subject to any disclosure and nowadays do not arouse much controversy (Farid, 2022; Geddes, 2021; Habgood-Coote, 2023), even though they may significantly influence the perception of the presented image. The factor that may qualitatively distinguish both situations is the use of AI that can reinforce technical feasibility, simplify and expand the scope of modifications, or even take over some part of interactions between creators and audiences (Farid, 2022), something that cannot be accomplished by traditional media and could only be achieved to a small extent by a real politician with decent acting skills. One may even plausibly argue that politicians can hire real actors as their doppelgangers, but it would be technically and legally debatable, and the depth of manipulation would still be limited by the actor's physical abilities. AI takes the capabilities of instant modification to a much higher level.

However, it should be emphasized that small changes based on retouching photos or refining audio recordings are not legally questionable and have become a permanent part of the general activities aimed at improving the quality of images or sounds that are widely accepted by society (Boté-Vericad & Vállez, 2022). We might have a sense of being slightly misled, but we somehow agree with this, as well as with the fact that politicians lie or wear a specific tie to subconsciously influence viewers' attitudes.

Our perceptions based on visualizations are then strongly embedded in the distortion. Filters, or automatic photo editing are currently automated and integrated with the standard devices. Even the common belief that "the camera never lies" can be effectively challenged for technical reasons (e.g., camera quality and selection of parameters), or the subjective choices made by the photographer adjusting the settings or reworking the outcome. This leads Share (2010) to the conclusion that "a photograph is always a decontextualized representation of reality". The epistemic disorder is accompanied by cognitive manipulations aimed at creating a specific vision of reality that is favorable to the creator or depicted subject. Smith (2024) claims that "most people are not aware of the fracturing of the relationship between the signifier and the signified", and behave as if this fracturing did not exist. Meanwhile, "the image has lost its status as an authoritative representation of reality" (Smith, 2024).

5. Disclaimers as a basic countermeasure

In the face of widely recognized threats, researchers consistently describe potential countermeasures against deep fakes that are mainly technology-oriented (authentication, watermarking, disclaimers to reveal the use of AI). The importance of strengthening social resilience in the form of media- and cyberliteracy is also underlined (Farid, 2022; Farid & Schindler, 2020; Habgood-Coote, 2023; Huijstee van et al., 2021), so the audience can independently and consciously decide how to react to the presented content. The mere disclosure of AI use does not prejudge the recipients' awareness of the possibility of manipulation (Łabuz & Nehring, 2024). However, it is a prerequisite, necessary information to consider the consequences of interacting with manipulated content.

Regulatory practice shows that disclosure is still a basic safeguard in the case of deep fakes, and so far the most prominent one in the EU Artificial Intelligence Act (European Commission, 2024a; Łabuz, 2023). Deployers are obliged to reveal the use of AI for creating deep fakes,

although the scope of labeling is still not set. However, this reveals certain legislative trends from which pragmatic conclusions should be drawn. The rationale behind transparency in the form of disclaimers is to allow the recipients to make informed choices or step back from a given situation by understanding the context, and it serves to enhance citizen's perception (European Commission, 2024a; Kim & Lee, 2019).

Informing the recipients about the use of AI to generate the content by adding proper disclosure is supposed to lower the degree of manipulation, and it touches upon the very basic perception of the content. The concept of disclosing the very fact of using political advertising is described similarly and is applied to ensure that citizens are able to make informed political choices without manipulation or coercion (European Commission, 2024b). As the addressees of these obligations are mainly legitimate actors, a high level of compliance can be expected.

Similar protections are introduced by regulatory frameworks in selected US states (Łabuz, 2025). There are also specific examples of such disclaimers, which, however, do not go beyond the basic information that AI was used to create the content. This is an important direction for implementing transparency obligations. Appropriate markings for AI-generated content also appear in social media. In this case, too, they come down to laconic information regarding the way the content was created. At the moment, transparency obligations do not go further, which may result in omitting a particularly important aspect of ontological and epistemic manipulation. Enhanced solutions might be adopted during the standardization process foreseen by the EU (European Commission, 2024), but their final shape will depend on the conceptualization of threats to democratic processes and the ethics of political advertising.

6. "Ship of Theseus" paradox and deep fakes political advertising

6.1. General overview

The regulatory frameworks must notice the complexity of threats. In this regard, the "Ship of Theseus" paradox may be helpful in portraying some challenges stemming from deep fakes political advertising. The conundrum originally refers to *diachronic identity* (the one that changes over time) (Scaltsas, 1980), but it might also be strongly linked with the notion of *likeness* (Bariach, Hogan, & McBride, 2024), which gives it an ontological dimension, on the one hand, and an epistemic dimension, on the other.

The paradox was initially introduced by Plutarch in the first century and refers to the gradual replacement of individual elements of the wooden Ship of Theseus with new ones (Wortmann & Mulder, 2016, pp. 299–306). Philosophers have asked the valid questions: Once the original parts are completely replaced, would the ship still be the same entity? Would it still be permissible to call it the "Ship of Theseus"? If not, at what point would it lose its native identity? Thomas Hobbes creatively developed the paradox in the seventeenth century by adding the issue of retaining the gradually replaced original ship's components and assembling a second ship from them (Adriaenssen & Alma, 2021). Ziemski (2018) indicates that the essence of the paradox is the problem of "identity, its continuity and constitutive elements of being, and finally – the distinction between essence and substrate and accidental elements and attributes". In this study, we highlight the role of substrates and attributes that collectively influence our perception of identity built on the digital visualizations. We analyze the integrity of identity and its perception, rather than an inherent metaphysical properties. Epistemic value is understood in terms of its function in truth adherence and reliable knowledge formation facing overlapping realities – the true one and the created one.

For many centuries, philosophers have had troubles finding answers to the above-mentioned questions. The lack of a clear solution to the paradox does not rule out searching for analogies in the digital world, which already has been done in relation to computational and linguistic science (Tripto et al., 2023). Scaltsas (1980), commenting on the paradox in 1980, pointed out that cases of conflict were rare in everyday life. Nowadays, however, they might have practical applications in regard to cyborgs, transhumanism, or the evolution of anthropomorphized algorithms (Geretto, 2024). Ziemski (2018) sees it as a thought experiment that should force us to analyze new problems, which is also important in the face of technological development.

The indicated emerging trends of deep fakes political advertising might add another dimension to the classic conundrum. In our approach to the paradox, we examine whether the digital visualization (A′) is a faithful representation of the original object (A), and whether particular modifications make it a visualization that represents a non-existent object with different features (we label this new representation as B'), which poses a problem of misattribution. One may wonder how many planks of the ship one needs to be replace before it loses its original identity. We wonder whether changes in the visualized features of a politician lead to the visualization of a non-existent person (ontological dimension), and whether an epistemically unprepared external observer can assess the consequences of these changes (epistemic dimension).

6.2. Ontological dimension

Usually, if there have been no alterations that objectively distort perception, we have no problems identifying the person depicted in the photo or video. We also have no grounds to question the veracity of the digital representation, even if our attachment to *ocularcentrism* may sometimes be deceptive (Geddes, 2021). In the case of faithful representations, in the form of non-modified photographs or recorded videos, even considering technical limitations and the abovementioned decontextualization (the camera indeed lies), the A′ is normally seen as a digital reflection of the A, and thus can be identified with the A. It presents a visualization of one's likeness and mirrors specific features. Therefore, it is justified to perceive it as the A′ that aims to visualize a fragment of the identity of the A.

But what if A′ is significantly modified? Would replacing certain personal features with different ones falsely mirror the object? In our opinion, it would result in creating new object B′ that has different characteristics, and thus does not faithfully mirror the A. Therefore, equipping politicians with features they do not naturally possess would result in identity distortion. It is not the A' anymore; the new digital object does not constitute a faithful representation of the A, and distorts the identity of the original object.

These considerations can call into question the general faithfulness of modern digital visualizations in which the image processing occurs. Filters, beautification, and removal of visual imperfections – due to all these changes, even an ordinary photograph may not be a faithful visualization. However, setting the limits of acceptability of processing seems technically unattainable, especially since standard photographs or recorded videos are widely considered to reflect reality. Therefore, as a society, we accept that we are being deceived to a small extent and collectively treat such visualizations as the A'.

Deep fakes can theoretically aim for visualizing likeness (Bariach, Hogan, & McBride, 2024). The representation of the real object with theoretically the same characteristics can be created, just like a standard photograph. However, deep fakes do not save a digital image of the object, but create it anew, mirroring specific characteristic features, behavior patterns, or personality traits. Should this digital representation still be perceived as the A'?

The original conceptualization of solving the "Ship of Theseus" paradox was based on the pretty simple story of replacing wooden parts of the ship, and it still brought a lot of controversy and contradictory solutions. Hobbes' undermined the identity of the ship after replacing some of its elements with new ones, perceiving it as a new object with different characteristics (Adriaenssen & Alma, 2021). In the spirit of paradox, we therefore focus on the differences between perpetuated reality and generated and modified reality.

According to Baudrillard (1995), such generated or modified objects can be described as *simulacra*, "something having merely the form or appearance of a certain thing, without possessing its substance or proper qualities" (Benson, 2013). Simulacra do not reflect their original models. Although we might be unable to distinguish simulacra from the original models, i.e., *a sign of reality* from *reality*, this sign is different from reality and does not constitute a faithful representation of the original model (Baudrillard, 1995).

This replacement creates a *hyperreality*, understood by Baudrillard (1995) as detached from referential reality and at the same time impossible to be distinguished from it. The copy, or *a sign of reality*, therefore creates its own reality. It applies "images and symbols that are detached from any underlying reality" or, in the framework of post-modernism, they "have no original or genuine counterpart" (Nosta, 2023). This, in turn, leads to the next stage of disruption of the value of objects, when simulation might even become "the dominant mode of representation" (Nosta, 2023).

AI undoubtedly provides unprecedented opportunities to create Baudrillard's hyperreality, which constitutes a simulated and extended world, not the real one. The audiences meeting with an AI-generated politician encounter with "models of a real without origin or reality" (Baudrillard, 1995, p. 3), and not with a faithful visualization of a real politician. Baudrillard draws attention to the detachment of representation from reality. Kalpokas (2020, p. 9) points to the same phenomenon that accompanies deep fakes, allowing for personalization of (perceived) reality and requiring the awareness of "the source of lived reality".

Capturing these relationships is easier in the case of deep fakes designed to directly interact with recipients. They can never be a faithful representation of the A, because they do not reflect physical properties of the A. AI-copy does not get tired, it can respond faster and more effectively. Or maintain multiple conversations at the same time. It possesses several features that the A cannot have. In the case of non-interactive creations, the difference comes down to inadequate representation of the model.

However, in all these cases, the new object B′ emerges that merely resembles the A, regardless of the scope of modification. It can mimic the *likeness* but is equipped with additional properties, and therefore it cannot be labeled as A'. A digital visualization of politicians cannot be identical to the real persons, but the modification of their visual and behavioral features leads to a reimagined identity that is functionally distinct. These considerations touch mostly upon the ontological dimension of changes, and in the face of technological development and algorithmizing of visualizations, they should become the subject of in-depth analyses on the ethical use of processing.

6.3. Epistemic dimension

Moreover, the object can be *perceived subjectively*, which refers to the epistemic value of the digital representation. Öhman (2022) analyzes the epistemic value of deep fakes in the case of non-consensual use of images. On the margin of his considerations, he refers to the "Ship of Theseus" paradox. He argues that "whether a person has a moral claim to content that allegedly uses their identity depends on the type of harm under consideration" (Öhman, 2022, p. 16). His research suggests that in some situations – depending on the context – the A may be identified with the B'. However, this identification is based on *likeness* or *resemblance* and applies primarily to the protection of an image infringed without the consent of the depicted person. The concern is not whether the modified entity is ontologically identical to the original, but whether it maintains sufficient visual and behavioral consistency to be misidentified as such (Öhman, 2022). Therefore, on the ontological level, the perpetuated representation of the identity of the depicted object is objectively disturbed, and on the epistemic level, it depends on the recipient's recognition and identification.

That is why it is so important to take into account the epistemic factor. The B′ is not the same as the A′, it does not faithfully reflect the properties of the A, but the audience's image of it may remain unchanged. An observer seeing a ship for the first time, without pre-information about its actual features, is epistemically unprepared to judge what is original and what is not. Objectively, the ship has changed. On a subjective level, it is still *perceived* as the same ship.

As a consequence, the ship's identity might be subjectively constructed anew during each interaction with recipients who do not know the ship's real properties. The integrity of the real image is objectively disturbed, but it is the *subjective* perception that determines the appropriate value and is used to draw conclusions about the identity of the depicted object. A similar phenomenon occurs in the case of interaction with the AI -representation of the politician. Attractive or not. Convincing or not. Electable or not. And "our uncritical and disproportionate reliance on technology-mediated sight" only enhances this threat (Geddes, 2021, p. 1043).

The false representation and inability to recognize that the B′ differs from the A′ and is not a faithful representation of the A can reinforce "in the public's conscious and subconscious mind inflated qualities, creating the illusion of attributes that the real person does not possess" (Pashentsev, 2023, p. 56). This is the basic problem of deep fakes political advertising that uses additional or changed properties of the depicted person. It refers to the difference between reality and its imagination. It also deepens the *simulacrum* phenomenon described by Baudrillard. The B′ does not faithfully mirror the identity of the A, it has different characteristics but a visible *likeness* or *resemblance*.

The definitions of deep fakes already take into account the deceptive nature of technology, stating that they *falsely appear to a person to be authentic or truthful*. These creations undermine fundamental notions of how we understand authenticity and how we imagine the digital mirroring of identity, even if the visualizations still preserve a basic likeness and resemblance.

7. Discussion and future directions

The above-described reasoning can be used to indicate possible consequences for potential regulations. If we legally allow for the possibility of interference in one's own image (and we do, as evidenced by the history of political advertising and our approach to photography), and technically cannot establish impassable limits (and we cannot, due to the infinite variety of possible modifications), their legal admissibility could be recognized even regardless of the problem of disturbed or partially false mirroring of identity through digital representations (Öhman, 2022). We are unable to say what features can be highlighted or altered and what cannot, and each time we would have to rely on pure discretion.

An alternative to allowing modifications would be to introduce a system of total bans. This is hard to imagine at the current level of technological development and its dissemination. It would also contradict the above-mentioned socially acceptable manipulations in the form of photo retouching. One may plausibly argue, whether a complete ban would violate the right to freely dispose of one's image. Discussions about setting legal limits on image disposal may soon intensify significantly. Currently, legislative initiatives focus on protecting copyrights to images that could be used non-consensually, as expressed, for example, by the Danish proposal from 2025 (Bryant, 2025). The issues we have raised, in turn, could open up a discussion about whether the right to freely change one's digital lookalike exists at all, and what its limitations might be. Our arguments are firmly rooted in protecting recipients from unconscious manipulation in political sphere.

The limits should therefore be seen not in setting the thresholds of permissible interference in the media, which in our opinion is technically and legally not feasible in the case of deep fakes political advertising, but in transparently informing recipients about the modifications made, in accordance with disclosure rules.

The very use of AI technology to create a new object based on the

original one means intervention in the technical and epistemic layer. Recipients shall receive a clear message that AI was used to create a specific material. The basic disclaimer that the object was created by AI (e.g., "This content was produced with the use of AI") informs the recipients that they are dealing with an artificial creation. This is the first barrier based on technological transparency. However, this general obligation is certainly based on the assumption that the recipient is aware, and has an adequate level of media literacy to understand the possible implications of AI-synthesis and its capabilities of creating manipulated representations. This is an incorrect assumption, given the low public awareness and subtlety of multi-level manipulation. Therefore, a second epistemic barrier is necessary to comprise the ontological and epistemic problems that would help overcome the disorders and sensitize the recipients.

It might be done within the framework of transparency obligations, especially if they are going to be a primary safeguard against deep fakes. They should be implemented into frameworks for political advertising. In our opinion, future disclaimers must clearly disclose: 1) that the material comprises political advertising, 2) that AI was used to generate it, 3) that it does not reflect the reality in its entirety, and 4) it poses the risk of manipulation. Due to the importance of transparency in political advertising, in order to additionally protect recipients, we assume that disclaimers explicitly informing that the presented features of the politician may not correspond to reality, or contain manipulation are needed as mandatory. Such presuppositions would be filled by an exemplary disclaimer: *Caution! You might be targeted by manipulation. This image/video/audio was generated by the artificial intelligence for the purpose of political advertising and it might manipulate or not reflect the real features of depicted persons, objects, places or other entities or events to falsely appear to be authentic or truthful.*

The regulatory solutions must be strongly anchored in what is achievable, particularly enhanced transparency obligations. The regulatory framework of the EU has not yet standardized the forms of disclaimers for deep fakes, but it might serve as a benchmark for future legislation, allowing for building more effective safeguards for society and democratic systems (Łabuz, 2024). This is undoubtedly a chance and opportunity to send a clear signal that deep fakes political advertising can be regulated, with care for public awareness and AI ethics. This is also an opportunity to reduce their manipulative potential, especially in those cases where the aim of the manipulation is not the good of an individual or group, or the spread of democratic values, but hiding what the manipulators find embarrassing or inconvenient in the struggle for political power (Soczyński, 2021).

It should be emphasized, however, that even far-reaching transparency obligations may be insufficient to completely counteract the long-term effects of exposure to false reality presented by deep fakes, or to protect recipients from falling into cognitive biases. Our attachment to disclaimers presented in this study is dictated by a common-sense approach based on the real and applicable solutions that are being currently tested within the regulatory frameworks. Research on their effectiveness is still being conducted, mainly in the context of disinformation; however, the scope of disclaimers is still limited and comes down to the basic labeling (e.g. Dobber et al., 2023). This area therefore requires specific empirical research on effectiveness of different forms of transparency obligations and how AI-human interactions are shaped by them. Our analysis can be a reference point for formulating extensive and diversified disclaimers.

We note that our approach may have significant limitations, and verification of the assumptions will be possible after conducting additional empirical research. However, the significant innovation of this study is the attention to different dimensions of manipulation using deep fakes for political advertising purposes. Another limitation could be seen in excessive trust in the interpretive abilities of recipients. The research of Ecker et al. (2022) showed, for example, a tendency to ignore warnings or resistance to correction. On the other hand, Lu and Yuan (2024) pointed out the effectiveness of warnings in relation to deep fakes, and Martel and Rand (2024) in relation to fact-checking labels, whereas Lockhart and Tessono (2025) noted that bigger labels (full--screen size) are more effective.

One should not disregard other important forms of safeguards, including strengthening media- and cyberliteracy. Extensive disclaimers are an element of building public awareness precisely because they comprehensively inform about various levels of manipulation with the use of deep fakes. However, in order to have an impact, they must be the basis for drawing conclusions about the value of the content by the audiences. Hence, for even the most sophisticated disclaimers to be able to operate at all, they must be met with appropriate cognitive resonance and a comprehensive understanding of the consequences of AI-based modifications. In an optimistic scenario, it could even help draw attention to the harmfulness of ongoing distortion of reality and promote authenticity as a value.

Safeguards could also be augmented with technological elements, which could include cryptographic solutions that confirm identity of a person recorded, or the lack of manipulation after a given image or video has been captured. Initiatives such as the Coalition for Content Provenance and Authenticity (C2PA), in which content is automatically digitally marked at the metadata level to reveal subsequent manipulation (Ayata, 2024), may offer hope for promoting authenticity, but their universal implementation is currently elusive. The ability to track the manipulation process may also be an answer to the technical limitations of disclosure itself. However, it does not solve the problem of entire synthesis, in which the image/video is created from scratch, and not modified based on an existing authentic content.

It may also turn out that the next stages of image or sound manipulation will be even more technologically advanced. One of the major concerns of the growing analytical and creative capabilities of generative AI is the ability to automatically produce content tailored to individual audiences (personalization) using data about recipients. AI-generated candidates might be able to "create highly customized versions of themselves to attract different demographics" (Farid, 2022, p. 9), and multiply the fake content spread online. These messages could possibly be tailored to our basic needs. AI Yoon or his later incarnations fed by data analysis, able to select a narrative and direct it to specific subgroups or even individuals, would open a new chapter of political manipulation. Creating AI and political advertising regulatory frameworks must therefore also take into account new phenomena and possible mismanagement of social network data that can be exploited to improve deep fake persuasion and jeopardize users' privacy (Cerruto et al., 2022).

Tailored messages strategy can be combined with the use of mimicry and facial resemblance. Olszanowski et al. (2022) proved the positive effect of blending facial features of targeted person's and the presented object in a way that excludes conscious perception of such synthesis, but allows subconscious self-identification. This, in turn, could translate into perceiving the object as more attractive (Bailenson et al., 2008).

Moreover, experiments with generating deep fakes that address recipients by name and use real-time translation to reach linguistic minorities were tested in India during the election campaign in 2024. They proved how another strategy can possibly be implemented to target local communities in this country (Butala et al., 2025; Shukla & Schneier, 2024).

8. Conclusions

Deep fakes could be seen as a doping that enhances political advertising but is not explicitly forbidden or even regulated at all. It raises legitimate ethical, philosophical, political, and legal concerns about the violation of freedom of choice based on awareness and authenticity of information, or further disorders of mirroring the identity by digital representations.

This article is the first attempt to comprehensively address the epistemic and ontological disruptions associated with the use of deep

fakes for political advertising purposes. Due to the innovative nature of the phenomenon, it does not provide answers to all questions. It also opens up new areas for discussion and research on the limits of digital manipulation, and perhaps even the right of an individual to freely present their own image using AI-powered alterations. Our analysis provides us with a variety of conclusions:

First (1), the line between true and false would be gradually diminishing, making real attributes and identity of politicians less important than in the past. The "Ship of Theseus" paradox, or its modified version, can be tested more frequently, and the *fleet of alternative political identities* of digital representations will be expanded. Manipulations of a political nature are an inseparable part of political reality, and the absolute sincerity of the candidates seems to be an unattainable delusion. Deep fakes political advertising develops existing patterns and uses the qualitative and quantitative potential of generative AI. It constitutes a significant threat to cumulative epistemic capabilities by presenting a *false reality* identified directly or indirectly with truth and referential reality, which is enhanced by reduced epistemic value of audio and visual content and difficulties in distinguishing what exactly constitutes truth.

Second (2), the "Ship of Theseus" paradox can be used to illustrate the difficulties in setting the limits of interference in the politicians' image. The paradox has many dimensions and does not have to be reduced solely to diachronic identity. Although over the centuries it has not been possible to find a clear answer to the questions posed by the paradox, it might help to visualize the ontological and epistemic problems associated with deep fakes political advertising. Deep fakes do not involve gradual transformation over time but rather the instant creation of modified representations that simulate reality. The common point, however, is the question of the extent of modification and the moment at which a visual representation ceases to be a visual representation and creates an entirely new reality. In our opinion, this happens automatically with the creation of a deep fake. What is important, however, is that such a new object can be identified by recipients with a real object, and that identification is normally based on resemblance or likeness. Similarly to Öhman (2022), we conclude that a visualization in the form of deep fakes enables at least partial mirroring of identity.

Third (3), the above-mentioned cases of the use of deep fakes political advertising, as understood in this study, may indicate emerging trends that are currently based on the desire to mobilize specific social groups (e.g., linguistic minorities or young people), but additionally allow for reducing existing barriers that prevent some communities from becoming more fully involved in democratic processes. Some of these applications may have a positive impact on society, but currently they should not be seen as the main goal of deep fakes political advertising. Moreover, the lack of an appropriate regulatory framework encourages opportunistic testing of the technology without taking into account the need to reduce its manipulative potential and sensitizing the recipients.

Fourth (4), transparency obligations for deep fakes are still not a global standard. Regulatory frameworks that are just being built must be a significant impulse for jurisdictions around the world, and take into account trends currently visible on a micro-scale in the realm of political advertising. Without indicating manipulation and informing about a new deeply faked identity that does not correspond with the real person's features, the effectiveness of basic disclaimers might turn out to be limited. Developing common standards that take into account the greatest possible sensitization of public opinion to the manipulative potential and inauthenticity of the new identities will not be an ideal solution, but it seems to be the most pragmatic one at the moment. We do not prejudge the effectiveness of this type of solutions, but for them to make sense at all, they must take into account the multi-level manipulation of deep fakes political advertising in the technical, ontological, and epistemic dimensions.

Fifth, (5), we believe that it will not be possible to completely ban deep fakes political advertising and apparently there is no such temptation on the regulatory side, especially if we, as a society, already accept different types of media manipulations. We ourselves contribute to this by creating our own *simulacra*, or more or less distorted reflections of reality. Therefore, a reasonable step is to come to terms with the inevitable, and admit that setting clear boundaries for such manipulations is impossible.

Sixth (6), it is regularly pointed out that one of the key ways to counteract sophisticated manipulation via deep fakes is significant strengthening of society's digital and analytical competences. Relying solely on technology and transparency would neglect an extremely important aspect – it is cognitive abilities, the ability to analyze and process information that serve as a protective barrier for society. Disclosure, even in its advanced form, provides a basis for inference, and inference provides a basis for assessing the actual value and veracity of the content.

Seventh (7), it will be important to pay attention to the further development of AI technology, which includes not only an increase in the quality of synthesis, but also the potential to use data to target individual recipients and personalize the messages. It will be advisable to integrate research on deep fakes with research on privacy-preserving data analytics, and promote the development of technologies and solutions that protect private users. Subsequently, this will serve to secure democratic processes against sophisticated forms of manipulation based on technology.

CRediT authorship contribution statement

Mateusz Łabuz: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Sławomir Soczyński:** Writing – original draft, Supervision, Conceptualization.

Declaration of competing interest

Both authors declare that there is no conflict of interest in regard to manuscript titled "Deep fakes as a tool of political advertising. Can regulatory framework benefit from the "Ship of Theseus" paradox?".

Data availability

No data was used for the research described in the article.

References

Abuelwafa, M. A. (2021). Legitimation and manipulation in political speeches: A corpus-based study. *Procedia Computer Science, 189*, 11–18. https://doi.org/10.1016/j.procs.2021.05.066

Adriaenssen, H. T., & Alma, S. (2021). Thomas Hobbes and Thomas White on identity and discontinuous existence. *Pacific Philosophical Quarterly, 102*, 429–454. https://doi.org/10.1111/papq.12354

Agarwal, A., & Ratha, N. (2023). Manipulating faces for identity theft via morphing and deepfake: Digital privacy. *Handbook of Statistics, 48*, 223–241. https://doi.org/10.1016/bs.host.2022.12.003

Ayata, O. (2024). *Artificial realities: Mitigations against deepfakes*. Centre for International Governance Innovation: Waterloo.

Bailenson, J. N., Iyengar, S., Yee, N., & Collins, N. A. (2008). Facial similarity between voters and candidates causes influence. *Public Opinion Quarterly, 72*(5), 935–961. https://doi.org/10.1093/poq/nfn064

Baudrillard, J. (1995). *Simulacra and simulation*. University of Michigan Press.

Benson, P. (2013). The ontology of photography: From analogue to digital. https://philosophynow.org/issues/95/The_Ontology_of_Photography_From_Analogue_To_Digital. (Accessed 15 January 2024).

Boté-Vericad, J.-J., & Vállez, M. (2022). Image and video manipulation: The generation of deepfakes. In P. Freixa, L. Codina, M. Pérez-Montoro, & J. Guallar (Eds.), *Visualisations and narratives in digital media. Methods and current trends* (pp. 116–127). https://doi.org/10.3145/indocs.2022.8. DigiDoc-EPI.

Brown, N. (2020). Deepfakes and the weaponization of disinformation. *Virginia Journal of Law and Technology, 23*(1).

Bryant, M. (2025). Denmark to tackle deepfakes by giving people copyright to their own features. http://theguardian.com/technology/2025/jun/27/deepfakes-denmark-copyright-law-artificial-intelligence. (Accessed 13 August 2025).

Butala, A., Nehring, C., & Łabuz, M. (2025). *The pervasive influence of A.I. on global political campaigns in 2024*. New Delhi: Future Shift Labs.

Cerruto, F., Cirillo, S., Desiato, D., Gambardella, S. M., & Polese, G. (2022). Social network data analysis to highlight privacy threats in sharing data. *Journal of Big Data, 9*, 1–26. https://doi.org/10.1186/s40537-022-00566-7

Chesney, B., & Citron, D. (2019). Deep fakes: A looming challenge for privacy, democracy, and national security. *California Law Review, 107*(18), 1753–1820.

Coppock, A., Hill, S., & Vavreck, L. (2020). The small effects of political advertising are small regardless of context, message, sender, or receiver: Evidence from 59 real-time randomized experiments. *Science Advances, 6*(36). https://doi.org/10.1126/sciadv.abc4046

Crain, M., & Nadler, A. (2019). Political manipulation and internet advertising infrastructure. *Journal of Information Policy, 9*, 370–410. https://doi.org/10.5325/jinfopoli.9.2019.0370

Dobber, T., Kruikemeier, S., Votta, F., Helberger, N., & Goodman, E. P. (2023). The effect of traffic light veracity labels on perceptions of political advertising source and message credibility on social media. *Journal of Information Technology & Politics*. https://doi.org/10.1080/19331681.2023.2224316

Dommett, K. (2019). The rise of online political advertising. *Political Insight, 10*(4), 12–15. https://doi.org/10.1177/2041905819891366

Ecker, U. K. H., Lewandowsky, S., Cook, J., Schmid, P., Fazio, L. K., Brashier, N., Kendeou, P., Vraga, E. K., & Amazeen, M. A. (2022). The psychological drivers of misinformation belief and its resistance to correction. *Nature Reviews Psychology, 1*, 13–29. https://doi.org/10.1038/s44159-021-00006-y

European Commission. (2024a). *Regulation (EU) 2024/1689 of the European Parliament and of the council of 13 June 2024 laying down harmonised rules on artificial intelligence*.

European Commission. (2024b). *Regulation (EU) 2024/900 of the European Parliament and of the council of 13 March 2024 on the transparency and targeting of political advertising*.

Europol. (2022). *Facing reality? Law enforcement and the challenge of deepfakes, an observatory report from the Europol Innovation Lab*. Publications Office of the European Union.

Fallis, D. (2021). The epistemic threat of deepfakes. *Philosophy & Technology, 34*(4), 623–643. https://doi.org/10.1007/s13347-020-00419-2

Farid, H. (2022). Creating, using, misusing, and detecting deep fakes. *Journal of Online Trust and Safety, 1*(4). https://doi.org/10.54501/jots.v1i4.56

Farid, H., & Schindler, H.-J. (2020). *Deep fakes. On the threat of deep fakes to democracy and society*. Konrad Adenauer Stiftung.

Fernandez, A. (2021). 'Deep fakes': Disentangling terms in the proposed EU artificial intelligence act. *UFITA Archiv für Medienrecht und Medienwissenschaft, 85*(2), 392–433. https://doi.org/10.5771/2568-9185-2021-2-392

Franz, M. M., & Ridout, T. N. (2010). Political advertising and persuasion in the 2004 and 2008 presidential elections. *American Politics Research, 38*(2), 303–329. https://doi.org/10.1177/1532673X09353507

Geddes, K. G. (2021). Ocularcentrism and deepfakes: Should seeing be believing? *Fordham Intellectual Property, Media and Entertainment Law Journal, 31*(4), 1042–1083.

Geretto, M. (2024). *Humans, angels, and cyborgs aboard Theseus' ship*. Cham: Palgrave Macmillan.

Gordon, J. (2024). The dawn of the AI election. Prospect. https://www.prospectmagazine.co.uk/politics/64396/the-dawn-of-the-ai-election. (Accessed 15 January 2024).

Habgood-Coote, J. (2023). Deepfakes and the epistemic apocalypse. *Synthese, 201*. https://doi.org/10.1007/s11229-023-04097-3

Halvarsson, M. (2024). *Deepfakes, real votes political parties' use of deepfakes through the lens of AI ethics and democracy theory*. Umea: Umea University.

Harris, K. R. (2024). Synthetic media detection, the wheel, and the burden of proof. *Philosophy & Technology, 37*(131), 1–20. https://doi.org/10.1007/s13347-024-00821-0

Harrison, M. (2023). Bizarre AI deepfake answer questions on behalf of presidential candidate. https://futurism.com/the-byte/presidential-campaign-ai-deepfake. (Accessed 15 January 2024).

Heinmaa, T. (2023). *Winning elections the right way*. International Institute for Democracy and Electoral Assistance. https://doi.org/10.31752/idea.2023.77

Heller, M. (1987). The best candidate approach to diachronic identity. *Australasian Journal of Philosophy, 65*(4), 434–451. https://doi.org/10.1080/00048408712343071

Home Security Heroes. (2023). AI deepfakes in 2024 election. https://www.homesecurityheroes.com/ai-deepfakes-in-2024-election. (Accessed 15 January 2024).

Huijstee van, M., van Boheemen, P., Das, D., Nierling, L., Jahnel, J., Karaboga, M., & Fatun, M. (2021). *Tackling deepfakes in European policy*. Brussels: European Parliamentary Research Service.

Izaguirre, A. (2023). Eric Adams' revelation that he uses AI to speak in Mandarin stirs outcry: 'The mayor is making deep fakes of himself'. https://fortune.com/2023/10/17/new-york-city-mnayor-eric-adams-uses-ai-to-speak-mandarin. (Accessed 15 January 2024).

Järvi, T., Mattila, M., & Schoultz, A. (2021). Finding the one: Ease of candidate choices in high information open-list PR systems. *Scandinavian Political Studies, 44*(4), 441–462. https://doi.org/10.1111/1467-9477.12210

Jaursch, J. (2020). Defining online political advertising. *Stiftung Neue Verantwortung: Berlin*.

Jeangene Vilmer, J.-B., Escorcia, A., Guillaume, M., & Herrera, J. (2018). *Information manipulation. A challenge for our democracies. Policy Planning Staff (CAPS) of the Ministry for Europe and Foreign Affairs and the Institute for Strategic Research (IRSEM) of the Ministry for the French Armed Force*. Paris.

Jones, K. (2021). Protecting political discourse from online manipulation: The international human rights law framework. *European Human Rights Law Review, 1*, 68–79.

Kahn, K. F., & Geer, J. G. (1994). Creating impressions: An experimental investigation of political advertising on television. *Political Behavior, 16*, 93–116.

Kalpokas, I. (2020). Problematising reality: The promises and perils of synthetic media. *SN Social Sciences, 1*(1), 1–11. https://doi.org/10.1007/s43545-020-00010-8

Kim, D. (2022). *Is Korea showing the future of elections: AI candidates*. KEI. https://keia.org/the-peninsula/is-korea-showing-the-future-of-elections-ai-candidates.

Kim, S., & Lee, J. (2019). Citizen participation, process, and transparency in local government: An exploratory study. *Policy Studies Journal, 47*(4), 1026–1047. https://doi.org/10.1111/psj.12236

Kleemann, A. (2023). *Deepfakes – Wenn wir unseren Augen und Ohren nicht mehr trauen können, 43*. SWP-Aktuell. https://doi.org/10.18449/2023A43

Koc-Michalska, K., Lilleke, D. G., & Vedel, T. (2016). Civic political engagement and social change in the new digital age. *New Media & Society, 18*(9), 1807–1816. https://doi.org/10.1177/1461444815616218

Krueger, N., Vananmala, M., & Dave, R. (2023). Recent advancements in the field of deepfake detection. *arXiv*. https://doi.org/10.48550/arXiv.2308.05563. https://arxiv.org/abs/2308.05563. (Accessed 15 January 2024)

Łabuz, M. (2023). Regulating deep fakes in the artificial intelligence act. *Applied Cybersecurity & Internet Governance, 2*(1). https://doi.org/10.60097/ACIG/162856

Łabuz, M. (2024). Deep fakes and the artificial intelligence act—An important signal or a missed opportunity? *Policy & Internet, 16*(24), 783–800. https://doi.org/10.1002/poi3.406

Łabuz, M. (2025). A teleological interpretation of the definition of deep fakes in the EU artificial intelligence act – A purpose-based approach to potential problems with the word "existing". *Policy & Internet*. https://doi.org/10.1002/poi3.435

Łabuz, M., Małecki, M., & Mika-Łabuz, K. (2025). Child pornography in the form of deep fakes – Potential problems under Polish criminal law and proposals for changes. *Prawo w Działaniu, 61*, 54–84. https://doi.org/10.32041/pwd.6103

Łabuz, M., & Nehring, C. (2024). On the way to deep fake democracy? Deep fakes in election campaigns in 2023. *European Political Science, 23*, 454–473. https://doi.org/10.1057/s41304-024-00482-9

Łabuz, M., & Strnad, G. (2025). Political advertising with the use of deep fakes – Can deliberative democracy really benefit from this strategy? *Politics & Policy, 53*(3). https://doi.org/10.1111/polp.70035

Langguth, J., Pogorelov, K., Brenner, S., Filkuková, P., & Schroeder, D. T. (2021). Don't trust your eyes: Image manipulation in the age of DeepFakes. *Frontiers in Communication, 6*. https://doi.org/10.3389/fcomm.2021.632317

Lockhart, A., & Tessono, C. (2025). *Human or AI? Evaluating labels on AI-Generated social media content*. Toronto Metropolitan University.

Lu, H., & Yuan, S. (2024). "I know it's a deepfake": The role of AI disclaimers and comprehension in the processing of deepfake parodies. *Journal of Communication, 74*(5), 359–373. https://doi.org/10.1093/joc/jqae022

Martel, C., & Rand, D. G. (2024). Fact-checker warning labels are effective even for those who distrust fact-checkers. *Nature Human Behaviour, 8*, 1957–1967. https://doi.org/10.1038/s41562-024-01973-x

Mills, C. (1995). Politics and manipulation. *Social Theory and Practice, 21*(1), 97–112.

Mirsky, Y., & Lee, W. (2021). The creation and detection of deepfakes: A survey. *ACM Computing Surveys, 54*(1), 1–41. https://doi.org/10.1145/3425780

Nagumotu, K. (2022). Deep fakes are taking over social media: Can the law keep up? *Intellectual Property Law Review, 62*(2), 102–146.

Neudert, L. M., & Marchal, N. (2019). Polarisation and the use of technology in political campaigns and communication. *Panel for the Future of Science and Technology: Brussels*.

Nightingale, S. J., & Farid, H. (2022). AI-synthesized faces are indistinguishable from real faces and more trustworthy. *PNAS, 119*(8). https://doi.org/10.1073/pnas.2120481119

Nilesh, C. (2020). We've just seen the first use of deepfakes in an Indian election campaign. https://www.vice.com/en/article/jgedjb/the-first-use-of-deepfakes-in-indian-election-by-bjp. (Accessed 14 January 2024).

Noggle, R. (2021). Manipulation in politics. In *Oxford research encyclopedia of politics*. https://doi.org/10.1093/acrefore/9780190228637.013.2012

Nosta, J. (2023). Postmodernism and the hyperreality of GPT. https://johnnosta.medium.com/postmodernism-and-the-hyperreality-of-gpt-7e4aa9e181fd. (Accessed 14 January 2024).

Öhman, C. (2022). The identification game: Deepfakes and the epistemic limits of identity. *Synthese, 200*, 1–19. https://doi.org/10.1007/s11229-022-03798-5

Olalekan, O., & Sam, S. (2018). Empirical and methodological analysis of political advertisements. *Review of Journalism and Mass Communication, 6*(1&2), 12–21. https://doi.org/10.15640/rjmc.v6n1-2a2

Olszanowski, M., Lewandowska, P., Ozimek, A., & Frankowska, N. (2022). The effect of facial self resemblance on emotional mimicry. *Journal of Nonverbal Behavior, 46*(2), 197–213. https://doi.org/10.1007/s10919-021-00395-x

Pashentsev, E. (2023). The malicious use of deepfakes against psychological security and political stability. In E. Pashentsev (Ed.), *The palgrave handbook of malicious use of AI and psychological security* (pp. 47–80). https://doi.org/10.1007/978-3-031-22552-9_3

Pawelec, M. (2022). Deepfakes als Chance für die Demokratie? In A. Bogner, M. Decker, M. Nentwich, & C. Scherz (Eds.), *Digitalisierung und die Zukunft der Demokratie - Beiträge aus der Technikfolgenabschätzung* (pp. 89–102). https://doi.org/10.5771/9783748928928-89

NSA, FBI, & CISA. (2023). Contextualizing deepfake threats to organizations. https://media.defense.gov/2023/Sep/12/2003298925/-1/-1/0/CSI-DEEPFAKE-THREATS.PDF. (Accessed 14 January 2024).

Martin, T. W., & Yoon, D. (2022). These campaigns hope 'Deepfake' candidates help get out the vote. https://www.wsj.com/articles/these-campaigns-hope-deepfake-candidates-help-get-out-the-vote-11646756345. (Accessed 15 January 2024).

Bariach, B., Hogan, B., & McBride, K. (2024). Faces of the future: How generative AI is redefining likeness and identity in the age of artificial intelligence. https://www.oii.ox.ac.uk/news-events/faces-of-the-future-how-generative-ai-is-redefining-likeness-and-identity-in-the-age-of-artificial-intelligence. (Accessed 15 February 2024).

Pawelec, M., & Łabuz, M. (2025). *Non-Consensual Sexualising Deepfakes – Threats and Recommendations for Legal and Societal Action*. Warsaw: CEE Digital Democracy Watch.

Rini, R. (2020). Deepfakes and the epistemic backstop. *Philosophers' Imprint, 20*(24).

Scaltsas, T. (1980). The ship of theseus. *Analysis, 40*(3), 152–157. https://doi.org/10.1093/analys/40.3.152

Share, J. (2010). *Camera always lies, the*. Center for Media Literacy. from https://www.medialit.org/reading-room/camera-always-lies. (Accessed 15 January 2024).

Shin, H., & Yi, H. Y. (2022). *South Korea candidates woo young voters with 'deepfakes,' hair insurance*. Reuters. Retrieved January 15, 2024, from https://www.reuters.com/world/asia-pacific/skorea-candidates-woo-young-voters-with-deepfakes-hair-insurance-2022-03-03.

Shukla, V., & Schneier, B. (2024). Indian election was awash in deepfakes – But AI was a net positive for democracy. https://theconversation.com/indian-election-was-awash-in-deepfakes-but-ai-was-a-net-positive-for-democracy-231795. (Accessed 15 January 2025).

Simon, S. (2023). *Opinion: Did he really say that?* NPR. Retrieved January 15, 2024, from https://www.npr.org/2023/10/21/1207749595/opinion-did-he-really-say-that.

Smith, N. (2024). Seeing is not believing: Reality, simulation and the epistemic authority of the image. https://thetechnosocial.com. (Accessed 15 January 2025).

Soczyński, S. (2021). (Re)definicja pojęcia tabu medialnego jako kategorii aksjonormatywnej z perspektywy teoretyczno-funkcjonalnej. *Petrus: Cracow*.

Susser, D., Roessler, B., & Nissenabaum, H. (2019). Technology, autonomy and manipulation. *Internet Policy Review, 8*(2), 1–22. https://doi.org/10.14763/2019.2.1410

Tanusondjaja, A., Michelon, A., Hartnett, N., & Stocchi, L. (2023). Reaching voters on social media: Planning political advertising on Snapchat. *International Journal of Market Research, 65*(5), 566–580. https://doi.org/10.1177/14707853231175085

Tripto, I. N., Venkatraman, S., Macko, D., Moro, R., Srba, I., Uchendu, A., Le, T., & Lee, D. (2023). A ship of theseus: Curious cases of paraphrasing in LLM-generated texts. https://arxiv.org/abs/2311.08374. (Accessed 15 November 2024). doi:10.48550/arXiv.2311.08374.

Twomey, J., Ching, D., Aylett, M. P., Quayle, M., Linehan, C., & Murphy, G. (2023). Do deepfake videos undermine our epistemic trust? A thematic analysis of tweets that discuss deepfakes in the Russian invasion of Ukraine. *PLoS One, 18*(10). https://doi.org/10.1371/journal.pone.0291668

Vaccari, C., & Chadwick, A. (2020). Deepfakes and disinformation: Exploring the impact of synthetic political video on deception, uncertainty, and trust in news. *Social Media + Society, 6*(1). https://doi.org/10.1177/2056305120903408

Vaccari, C., & Valeriani, A. (2021). *Outside the bubble: Social media and political participation in Western democracies*. Oxford Academic. https://doi.org/10.1093/oso/9780190858476.001.0001

Vazquez Llorente, R. (2024). Deepfakes in the dock: Preparing international justice for generative AI. *The SciTech Lawyer, 20*(2), 29–33.

Walczyna, T., & Piotrowski, Z. (2023). Quick overview of face swap deep fakes. *Applied Sciences, 13*. https://doi.org/10.3390/app13116711

Whitfield, G. (2022). On the concept of political manipulation. *European Journal of Political Theory, 21*(4). https://doi.org/10.1177/14748851209322

Wortmann, H., & Mulder, W. (2016). Reflections on identity management in smart industry: The paradox of theseus' ship and beyond. *IFIP international conference on advances in production management systems (APMS)*. https://doi.org/10.1007/978-3-319-51133-7_36

Yadlin-Segal, A., & Oppenheim, Y. (2021). Whose dystopia is it anyway? Deepfakes and social media regulation. *Convergence: The International Journal of Research Into New Media Technologies, 27*(1), 36–51. https://doi.org/10.1177/1354856520923963

Zhuravskaya, E., Petrova, M., & Enikolopov, R. (2020). Political effects of the internet and social media. *Annual Review of Economics, 12*, 415–438. https://doi.org/10.1146/annurev-economics081919-050239

Ziemski, M. (2018). Paradoks statku Tezeusza, czyli o relacji prawa i filozofii. https://ingremio.org/felieton/paradoks-statku-tezeusza-czyli-o-relacji-prawa-i-filozofii. (Accessed 15 January 2024).

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